

# H43 Autocorrelation Momentum

Intraday momentum on NQ futures, taken only when recent 5-minute returns have been following through (positive autocorrelation): research report with fixed-risk sizing (cohort member #12, C16cd8e3c605af462)

Prepared for the investment committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract (\$20/point) unless stated

**Recommendation: FOURTH CANDIDATE: FORWARD-PAPER ONLY FOR NOW. Do not add it to the evaluation account.** It has the strongest search-corrected confirmation result of the four candidates (t 3.27, family-wise p 0.020), and it has been positive in R every year since 2018, including every year after the freeze. But its untouched holdout was only marginal (PF 1.14, t 1.15, null p 0.079), and 2026 lost money on fixed contracts. It is the same kind of trade as H41 (holdout daily correlation 0.40). In the simulations, adding it to an H41 + H47 + H25 account *lowered* the pass rate.

## Executive summary

- **Why this is the fourth candidate.** After H41, H47 and H25 short, the next names on post-freeze evidence were H20 (lost \$52.6k in 2026, correlation 0.60 with H41), #22 (a copy of H47, correlation 0.59) and H09 short. H09's profit exists only on fixed contracts: in R it made −15R over 2010–26 and +0.01R per trade in confirmation, so it does not survive fixed-risk sizing. H43 is positive in R (+0.12R per trade over 4,094 trades), so it is the best remaining choice for a fixed-risk trader.
- **What it is.** It buys (or sells short) when the 15-minute move first crosses +1 (or −1) standard deviation, but only if the last 24 five-minute returns show positive autocorrelation ( $\geq 0.25$ ): moves have been following through, not reversing. The stop is  $1 \times \text{ATR}$ . No target, flat at 16:00. **About 277 trades a year, roughly 5 a week and up to 7-9 on a busy day.**
- **Evidence ladder.** Discovery 2010–18: +\$37.0k, 2,123 trades, PF 1.16, t 1.88, but essentially zero in R (+0.03R per trade). Confirmation 2019–22: +\$156.6k, 970 trades, PF 1.39, t 3.27, null p 0.020, family-wise p 0.020, tier PROMISING. **Holdout 2023–25: +\$65.0k, 836 trades, PF 1.14, t 1.15, null p 0.079, SUPPORTED.** 2026: −\$2.6k (PF 0.98), INCONCLUSIVE, yet +26R per unit of risk.
- **After the freeze** (2023–Aug 2026): 1,001 trades, PF 1.10, +\$62.3k on 1 NQ, +117R, +0.12R per trade (t 1.55 in R). Every year was positive in R: +37R, +20R, +34R, +26R.
- **Main weaknesses.** (1) The holdout was marginal and well below confirmation (z −1.24). (2) The autocorrelation filter, the core of the idea, stopped adding value after the freeze: unfiltered momentum did about as well (holdout PF 1.11 vs 1.14). (3) Like H41, it made nothing in 2010–17 (−57R, a 6-year drawdown). (4) The edge is concentrated in longs, first-hour entries and calm days. (5) It overlaps H41.
- **Fixed risk.** It improves the profit factor (2019–26: about 1.22 → 1.28). At \$150 it needs H41's '1 MNQ up to \$250' rule, because 1 MNQ now risks about \$120. As a fourth sleeve it reduced simulated evaluation pass rates, so keep it on paper.

## Headline results by research stage

| Period                  | Trades | Per week | Net P&L   | PF   | Win % | Net R  | Avg R  | Max DD   | Verdict      |
|-------------------------|--------|----------|-----------|------|-------|--------|--------|----------|--------------|
| Discovery 2010–18       | 2,123  | 4.7      | \$37,013  | 1.16 | 27.9% | +68.7  | +0.032 | \$9,450  | PRIMARY      |
| Confirmation 2019–22    | 970    | 4.7      | \$156,625 | 1.39 | 30.5% | +314.3 | +0.324 | \$28,955 | PROMISING    |
| Holdout 2023–25         | 836    | 5.3      | \$64,961  | 1.14 | 27.6% | +90.2  | +0.108 | \$31,523 | SUPPORTED    |
| 2026 audit (Jan–Aug)    | 165    | 5.1      | −\$2,630  | 0.98 | 33.3% | +26.4  | +0.160 | \$32,482 | INCONCLUSIVE |
| 2024–2025 only          | 572    | 5.5      | \$44,732  | 1.12 | 25.9% | +53.6  | +0.094 | \$28,156 | –            |
| All post-freeze 2023–26 | 1,001  | 5.3      | \$62,331  | 1.10 | 28.6% | +116.6 | +0.117 | \$32,482 | –            |

*Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. R = P&L divided by the trade's own initial risk.*

# 1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

| Stage               | Dates                   | Role                              | What could change                     |
|---------------------|-------------------------|-----------------------------------|---------------------------------------|
| Discovery           | 2010-06-07 → 2018-12-31 | search (in-sample)                | everything: rules, grid, management   |
| Confirmation        | 2019 → 2022             | out-of-sample test vs null worlds | only which frozen candidates are kept |
| Final cohort freeze | 22 Sep 2026 23:21 UTC   | spec hash f5666dac...             | nothing after this point              |
| Holdout             | 2023 → 2025             | untouched out-of-sample           | nothing (report only)                 |
| 2026 audit          | 2026-01-01 → 2026-08-10 | secondary, low power              | nothing (report only)                 |
| True forward        | from 2026-09-24         | live-market paper record          | nothing (forward/run_forward.py)      |

**Reproducibility.** For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

**Why R-multiples matter here.** NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R (P&L ÷ initial risk), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

## Hypothesis family

**Preregistered hypothesis (H43).** Momentum  $Z(15) \geq z$ , with a confluence condition: the lag-1 autocorrelation of the last 24 five-minute returns (two hours) is at least  $a$ . Grid:  $z \in \{1.0, 1.5\}$ ,  $a \in \{\text{none}, 0.1, 0.25\}$ , continuation, long / short / both. In discovery 458 specifications were tested, 23 were PRIMARY, and the best  $t$  was 2.51.

**Rationale.** Positive autocorrelation means one 5-minute move has tended to be followed by another in the same direction: order flow is persistent, not mean-reverting. A strong 15-minute push inside that regime should carry further than the same push in a choppy market. This is the same idea as H41, measured with autocorrelation instead of a variance ratio. That is why the two overlap.

**The filter's record.** In discovery the autocorrelation condition raised the average trade in 92% of 220 paired tests, and in 85% of 41 pairs in confirmation. For this exact entry ( $z$  1.0, both directions) the filtered version beat the unfiltered one clearly before the freeze (PF 1.16 vs 0.93 in discovery, 1.39 vs 1.05 in confirmation). After it, the gap closed: 1.14 vs 1.11 in the holdout, 0.98 vs 1.06 in 2026.

**Behaviour cluster.** Cluster #41: PROMISING, 17 confirmation survivors, 4 entry rules. Holdout daily correlation is 0.50 with H20 and 0.40 with H41, the two other momentum members.

## 2. H43 (#12, long + short): specification

| Field                            | Frozen value                                                                                                                                                           |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategy ID                      | C16cd8e3c605af462                                                                                                                                                      |
| Entry rule ID                    | E4de4e4fa7bbaa856                                                                                                                                                      |
| Family / version                 | v3_h43_pos_autocorr_momentum (v3.0)                                                                                                                                    |
| Instrument                       | NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars                                                                                          |
| Direction                        | both                                                                                                                                                                   |
| Mode                             | continuation                                                                                                                                                           |
| Long trigger                     | the 15-minute move first exceeds +1.0 sigma units with the 5-minute return autocorrelation (last 24) $\geq 0.25$                                                       |
| Short trigger                    | the 15-minute move first falls below -1.0 sigma units with the 5-minute return autocorrelation (last 24) $\geq 0.25$                                                   |
| Parameters                       | {"a": 0.25, "mode": "continuation", "z": 1.0}                                                                                                                          |
| Signal timing                    | evaluated at the close of each 1-minute bar                                                                                                                            |
| Entry order                      | market order at the open of the next 1-minute bar (same session and contract)                                                                                          |
| Entry window                     | 09:30–15:30 New York; one position at a time; no pyramiding                                                                                                            |
| Initial stop                     | 1.0 $\times$ ATRh from entry (ATRh = ATR(14) of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick |
| Target                           | none (100% of the position runs)                                                                                                                                       |
| Break-even / trailing / partials | none / none / none                                                                                                                                                     |
| Time exit                        | flat at the close of the last bar before 16:00 New York                                                                                                                |
| Same-bar ambiguity               | pessimistic (stop assumed first)                                                                                                                                       |
| Costs (baseline)                 | \$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26                                                                       |
| Cost file version                | costs_v2_approved_2026-09-21#a27ed5df0c04                                                                                                                              |

### Statistical evidence trail

| Stage                | Test / statistic                                         | Result                       | Verdict            |
|----------------------|----------------------------------------------------------|------------------------------|--------------------|
| Discovery 2010–18    | label (trades $\geq 30$ , net $> 0$ , t $\geq 1$ )       | 2123 trades, PF 1.16, t 1.88 | PRIMARY            |
| Discovery            | parameter robustness (share of grid neighbours positive) | 1.00                         |                    |
| Discovery            | sub-periods positive (of 3) / years positive (of 9)      | 1 / 4                        |                    |
| Confirmation 2019–22 | t-stat of mean trade                                     | 3.27                         |                    |
| Confirmation         | null p (zero-drift worlds) / (drift worlds) / max        | 0.020 / 0.020 / 0.020        | $p \leq 0.05$      |
| Confirmation         | family-wise p (max-t in same frequency bucket)           | 0.020                        | significant at 10% |
| Confirmation         | parameter support / management support                   | 0.50 / 0.46                  |                    |
| Confirmation         | years positive (of 4); stress net                        | 3; \$144,985                 |                    |
| Confirmation         | evidence tier (preregistered)                            |                              | PROMISING          |
| Behaviour cluster    | cluster at p 0.50 (members / entry rules)                | #41 (17 / 4)                 | PROMISING          |
| Holdout 2023–25      | t-stat; z vs confirmation mean trade                     | 1.15; -1.24                  |                    |
| Holdout              | null p (100 holdout null worlds)                         | 0.079                        |                    |
| Holdout              | classification (preregistered)                           |                              | SUPPORTED          |
| 2026 audit           | t-stat; net                                              | -0.09; -\$2,630              | INCONCLUSIVE       |

*Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.*

## H43 (#12, long + short): how it trades

**Signal (checked at every 1-minute bar close, 09:30-15:30 New York).** Two conditions must hold. (1) The 15-minute move, in sigma units (the move divided by that time of day's normal 15-minute volatility), *first* crosses +1.0 (long) or -1.0 (short). (2) The autocorrelation of the last 24 five-minute returns is at least 0.25. When both hold, it trades at market on the next bar. It holds one position at a time and can re-enter later in the day.

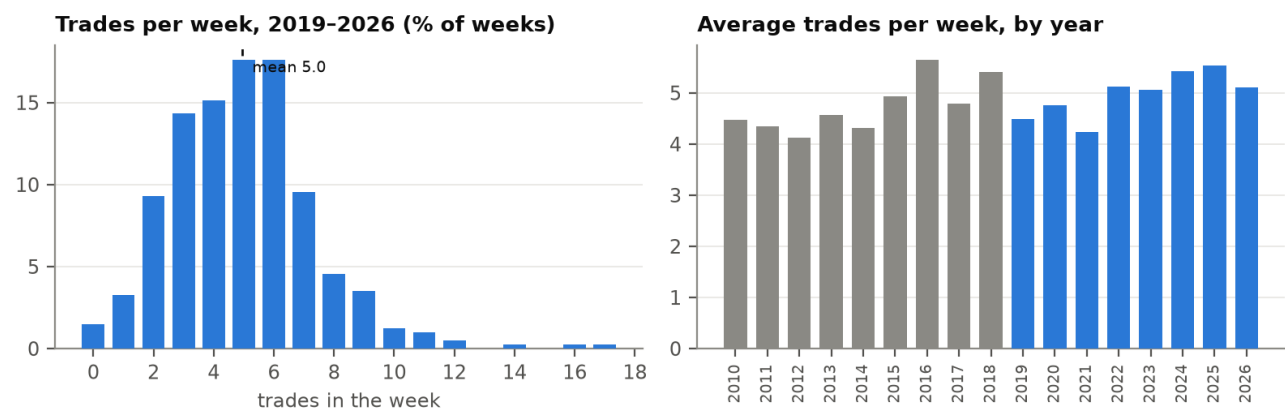
**Risk and exit.** The stop sits  $1 \times \text{ATR}_h$  from the entry ( $\text{ATR}(14)$  of completed 15-minute bars): about 31-47 NQ points in 2023-25 and 60 in 2026. One NQ risked about \$1,290 per trade in 2026; one MNQ about \$120-130. There is no target, no trailing stop and no break-even. It is flat at the last bar before 16:00.

**What a trade looks like.** 68% of trades are stopped out (median 23 minutes, average  $-1.09R$ ). The 32% held to the close averaged  $+2.65R$  (median 230 minutes). The win rate is 29%, and the average winner is about  $3 \times$  the average loser. Entries between 09:30 and 10:00 averaged  $+0.33R$  ( $+0.47R$  after the freeze). Later entries were about zero. This is descriptive only.



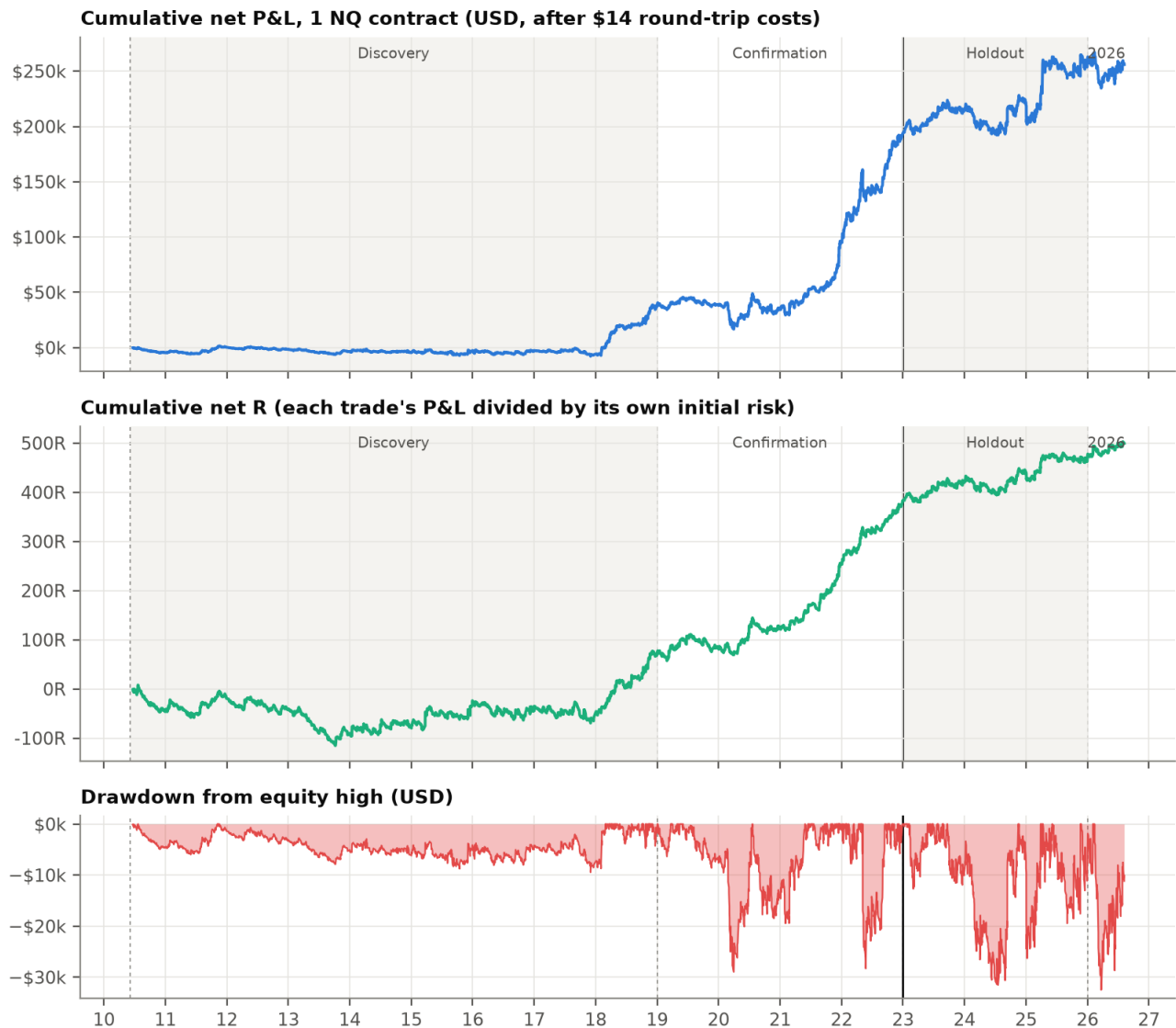
Real holdout trades, 1-minute NQ bars (grey band = bar range, line = close). Blue dotted = entry price, red dashed = the fixed stop; nothing else is managed.

### How often it trades



| Frequency (2019-Aug 2026)   | Value     |
|-----------------------------|-----------|
| Average trades per week     | 5.0       |
| Median trades per week      | 5         |
| Weeks with no trade         | 1.5%      |
| Weeks with exactly 1 trade  | 3.3%      |
| Weeks with 2 or more trades | 95.2%     |
| Busiest week                | 17 trades |
| Longest gap without a trade | 1 weeks   |
| Average trades per year     | 259       |

## H43 (#12, long + short): performance



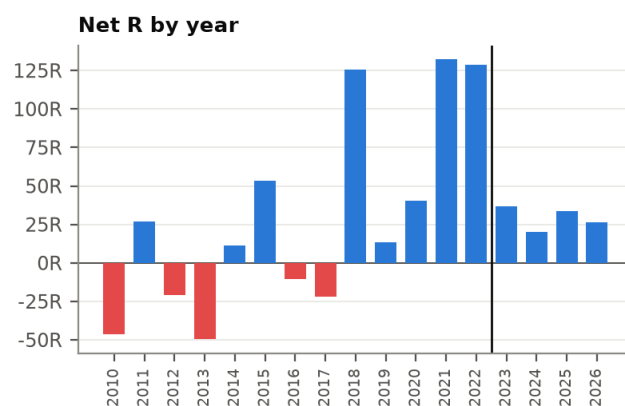
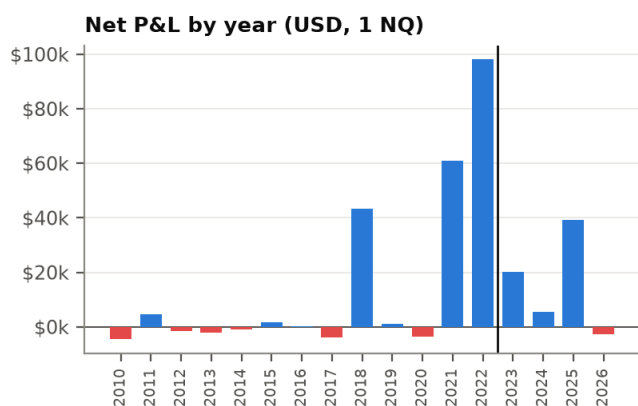
Shaded bands mark the research stages; the solid black line is the start of data never seen before the freeze (1 Jan 2023).

### H43 (#12, long + short): full statistics by period

| Metric                            | Discovery<br>2010-18  | Confirm.<br>2019-22    | Holdout<br>2023-25      | 2026<br>Jan-Aug       | 2024-25                 | Post-freeze<br>2023-26  | Full<br>2010-26         |
|-----------------------------------|-----------------------|------------------------|-------------------------|-----------------------|-------------------------|-------------------------|-------------------------|
| Trades                            | 2,123                 | 970                    | 836                     | 165                   | 572                     | 1,001                   | 4,094                   |
| Trades / year                     | 248                   | 243                    | 279                     | 268                   | 286                     | 277                     | 253                     |
| Trades / week (all weeks)         | 4.7                   | 4.7                    | 5.3                     | 5.1                   | 5.5                     | 5.3                     | 4.8                     |
| Weeks with at least 1 trade       | 98%                   | 98%                    | 100%                    | 100%                  | 100%                    | 100%                    | 98%                     |
| Trades / week (active weeks)      | 4.8                   | 4.8                    | 5.3                     | 5.2                   | 5.4                     | 5.3                     | 4.9                     |
| Days with a trade (of sessions)   | 62%                   | 62%                    | 65%                     | 70%                   | 66%                     | 66%                     | 63%                     |
| Max trades in one day             | 7                     | 9                      | 7                       | 4                     | 7                       | 7                       | 9                       |
| Long share                        | 48%                   | 49%                    | 48%                     | 45%                   | 47%                     | 47%                     | 48%                     |
| Net P&L (baseline \$14)           | \$37,013              | \$156,625              | \$64,961                | −\$2,630              | \$44,732                | \$62,331                | \$255,969               |
| Gross P&L (no costs)              | \$66,735              | \$170,205              | \$76,665                | −\$320                | \$52,740                | \$76,345                | \$313,285               |
| Net P&L, moderate \$9             | \$47,628              | \$161,475              | \$69,141                | −\$1,805              | \$47,592                | \$67,336                | \$276,439               |
| Net P&L, stress \$26              | \$11,537              | \$144,985              | \$54,929                | −\$4,610              | \$37,868                | \$50,319                | \$206,841               |
| Profit factor (net)               | 1.16                  | 1.39                   | 1.14                    | 0.98                  | 1.12                    | 1.10                    | 1.21                    |
| Profit factor (gross)             | 1.31                  | 1.43                   | 1.17                    | 1.00                  | 1.15                    | 1.13                    | 1.26                    |
| Profit factor (stress)            | 1.05                  | 1.35                   | 1.11                    | 0.97                  | 1.10                    | 1.08                    | 1.16                    |
| Win rate                          | 27.9%                 | 30.5%                  | 27.6%                   | 33.3%                 | 25.9%                   | 28.6%                   | 28.7%                   |
| Average trade                     | \$17                  | \$161                  | \$78                    | −\$16                 | \$78                    | \$62                    | \$63                    |
| Average winner                    | \$460                 | \$1,893                | \$2,323                 | \$2,441               | \$2,746                 | \$2,346                 | \$1,280                 |
| Average loser                     | −\$154                | −\$599                 | −\$780                  | −\$1,244              | −\$853                  | −\$851                  | −\$427                  |
| Payoff ratio (avg W / avg L)      | 2.99                  | 3.16                   | 2.98                    | 1.96                  | 3.22                    | 2.76                    | 2.99                    |
| Expectancy (net R / trade)        | +0.032                | +0.324                 | +0.108                  | +0.160                | +0.094                  | +0.117                  | +0.122                  |
| Total net R                       | +68.7                 | +314.3                 | +90.2                   | +26.4                 | +53.6                   | +116.6                  | +499.6                  |
| Median trade (R)                  | −1.08                 | −1.02                  | −1.01                   | −1.01                 | −1.01                   | −1.01                   | −1.02                   |
| t-stat of mean trade (USD)        | 1.88                  | 3.27                   | 1.15                    | −0.09                 | 0.85                    | 0.99                    | 3.14                    |
| t-stat of mean trade (R)          | 0.60                  | 3.79                   | 1.31                    | 0.87                  | 0.91                    | 1.55                    | 3.11                    |
| Sharpe, daily, annualised (USD)   | 0.63                  | 1.56                   | 0.66                    | −0.12                 | 0.59                    | 0.51                    | 0.76                    |
| Sharpe, daily, annualised (R)     | 0.20                  | 1.88                   | 0.74                    | 1.09                  | 0.63                    | 0.80                    | 0.76                    |
| Sortino (USD)                     | 1.25                  | 2.74                   | 1.26                    | −0.19                 | 1.15                    | 0.95                    | 1.39                    |
| Max drawdown (USD)                | \$9,450               | \$28,955               | \$31,523                | \$32,482              | \$28,156                | \$32,482                | \$32,482                |
| Max drawdown (R)                  | 123.3                 | 41.9                   | 38.8                    | 19.9                  | 38.8                    | 38.8                    | 123.3                   |
| Longest drawdown (calendar days)  | 2,295                 | 410                    | 417                     | 177                   | 308                     | 417                     | 2,295                   |
| Calmar (annual net / max DD)      | 0.46                  | 1.35                   | 0.69                    | −0.13                 | 0.79                    | 0.53                    | 0.49                    |
| Longest losing streak (trades)    | 19                    | 19                     | 19                      | 12                    | 19                      | 19                      | 19                      |
| Longest winning streak            | 6                     | 7                      | 5                       | 3                     | 3                       | 5                       | 7                       |
| Best trade                        | \$4,841               | \$8,526                | \$19,191                | \$8,261               | \$19,191                | \$19,191                | \$19,191                |
| Worst trade                       | −\$974                | −\$2,574               | −\$3,554                | −\$3,209              | −\$3,554                | −\$3,554                | −\$3,554                |
| Best day / worst day              | \$4,841 /<br>−\$4,268 | \$8,526 /<br>−\$16,916 | \$19,191 /<br>−\$11,148 | \$8,261 /<br>−\$5,182 | \$19,191 /<br>−\$11,148 | \$19,191 /<br>−\$11,148 | \$19,191 /<br>−\$16,916 |
| Top-10 trades as % of net         | 93%                   | 48%                    | 169%                    | −                     | 246%                    | 177%                    | 43%                     |
| Positive months                   | 47% of 103            | 58% of 48              | 61% of 36               | 25% of 8              | 54% of 24               | 55% of 44               | 51% of 195              |
| Average hold (minutes)            | 106                   | 114                    | 103                     | 114                   | 102                     | 105                     | 108                     |
| Avg initial risk (NQ points)      | 7.7                   | 31.7                   | 41.4                    | 64.6                  | 45.4                    | 45.3                    | 22.6                    |
| Avg initial risk, 1 NQ (USD) = 1R | \$154                 | \$634                  | \$829                   | \$1,293               | \$908                   | \$905                   | \$451                   |
| Average NQ price at entry         | 4,167                 | 11,281                 | 18,831                  | 27,118                | 20,913                  | 20,197                  | 9,772                   |

Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

## H43 (#12, long + short): calendar



| Year | Period  | N   | Net P&L  | Gross     | Stress   | PF   | Win % | Avg trade | Net R  | Avg R  | Max DD   | 1R (USD) |
|------|---------|-----|----------|-----------|----------|------|-------|-----------|--------|--------|----------|----------|
| 2010 | Disc.   | 134 | -\$4,476 | -\$2,600  | -\$6,084 | 0.57 | 22%   | -\$33     | -46.2  | -0.345 | \$4,937  | \$88     |
| 2011 | Disc.   | 227 | \$4,717  | \$7,895   | \$1,993  | 1.27 | 33%   | \$21      | +26.9  | +0.118 | \$2,796  | \$115    |
| 2012 | Disc.   | 215 | -\$1,635 | \$1,375   | -\$4,215 | 0.90 | 30%   | -\$8      | -21.1  | -0.098 | \$3,190  | \$102    |
| 2013 | Disc.   | 238 | -\$2,037 | \$1,295   | -\$4,893 | 0.87 | 27%   | -\$9      | -49.5  | -0.208 | \$5,126  | \$88     |
| 2014 | Disc.   | 225 | -\$1,025 | \$2,125   | -\$3,725 | 0.95 | 24%   | -\$5      | +11.5  | +0.051 | \$3,192  | \$125    |
| 2015 | Disc.   | 257 | \$1,762  | \$5,360   | -\$1,322 | 1.05 | 30%   | \$7       | +53.6  | +0.209 | \$5,342  | \$178    |
| 2016 | Disc.   | 295 | \$125    | \$4,255   | -\$3,415 | 1.00 | 27%   | \$0       | -10.3  | -0.035 | \$4,316  | \$154    |
| 2017 | Disc.   | 250 | -\$3,745 | -\$245    | -\$6,745 | 0.86 | 24%   | -\$15     | -22.0  | -0.088 | \$7,425  | \$134    |
| 2018 | Disc.   | 282 | \$43,327 | \$47,275  | \$39,943 | 1.71 | 31%   | \$154     | +125.8 | +0.446 | \$4,729  | \$329    |
| 2019 | Conf.   | 234 | \$1,169  | \$4,445   | -\$1,639 | 1.02 | 25%   | \$5       | +13.2  | +0.056 | \$8,592  | \$278    |
| 2020 | Conf.   | 248 | -\$3,652 | -\$180    | -\$6,628 | 0.97 | 27%   | -\$15     | +40.4  | +0.163 | \$24,099 | \$662    |
| 2021 | Conf.   | 221 | \$60,956 | \$64,050  | \$58,304 | 1.76 | 33%   | \$276     | +132.3 | +0.598 | \$8,609  | \$577    |
| 2022 | Conf.   | 267 | \$98,152 | \$101,890 | \$94,948 | 1.63 | 37%   | \$368     | +128.5 | +0.481 | \$28,292 | \$968    |
| 2023 | Holdout | 264 | \$20,229 | \$23,925  | \$17,061 | 1.18 | 31%   | \$77      | +36.6  | +0.139 | \$12,686 | \$657    |
| 2024 | Holdout | 283 | \$5,598  | \$9,560   | \$2,202  | 1.04 | 25%   | \$20      | +20.1  | +0.071 | \$28,156 | \$751    |
| 2025 | Holdout | 289 | \$39,134 | \$43,180  | \$35,666 | 1.19 | 27%   | \$135     | +33.5  | +0.116 | \$19,447 | \$1,062  |
| 2026 | Audit   | 165 | -\$2,630 | -\$320    | -\$4,610 | 0.98 | 33%   | -\$16     | +26.4  | +0.160 | \$32,482 | \$1,293  |

**Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)**

| Year | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul  | Aug   | Sep   | Oct   | Nov   | Dec   |
|------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|-------|-------|
| 2019 | +4.2  | -14.1 | +29.9 | -1.8  | +13.7 | +5.0  | -1.8 | -3.7  | -9.3  | -3.1  | -9.3  | +3.6  |
| 2020 | -8.2  | +12.6 | -16.9 | +21.8 | +3.1  | +34.1 | +4.5 | -6.2  | -7.6  | +3.2  | -0.1  | -0.0  |
| 2021 | -0.6  | +7.3  | +8.9  | +2.1  | +20.0 | +11.2 | -4.5 | +25.9 | -0.7  | +13.4 | +14.9 | +34.4 |
| 2022 | +27.5 | -8.9  | +6.6  | +41.9 | +1.4  | +5.3  | +1.0 | -0.6  | +14.9 | +10.2 | +15.7 | +13.4 |
| 2023 | +13.5 | -11.4 | -2.4  | +9.6  | +7.3  | +12.0 | +7.3 | -1.9  | +5.5  | -11.2 | +11.9 | -3.6  |
| 2024 | +6.9  | -18.2 | -0.8  | +6.6  | -10.6 | -5.5  | +6.5 | -1.5  | +19.9 | +2.3  | +17.8 | -3.2  |
| 2025 | -10.1 | +13.0 | +18.1 | +12.5 | +3.3  | -3.8  | +0.0 | -3.0  | -2.4  | -1.4  | +4.6  | +2.6  |
| 2026 | +1.3  | +13.0 | -4.5  | -3.0  | +15.9 | +4.7  | +1.5 | -2.5  |       |       |       |       |

## H43 (#12, long + short): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

## H43 (#12, long + short): breakdowns (all periods vs post-freeze 2023-26)

### Direction

| Direction | Trades (all) | Avg R (all) | PF (all) | Net (all) | Trades 23-26 | Avg R 23-26 | PF 23-26 | Net 23-26 |
|-----------|--------------|-------------|----------|-----------|--------------|-------------|----------|-----------|
| Long      | 1983         | +0.198      | 1.34     | \$193,238 | 474          | +0.201      | 1.33     | \$85,484  |
| Short     | 2111         | +0.051      | 1.09     | \$62,731  | 527          | +0.040      | 0.93     | -\$23,153 |

### Exit reason

| Exit          | Trades (all) | Avg R (all) | PF (all) | Net (all)    | Trades 23-26 | Avg R 23-26 | PF 23-26 | Net 23-26  |
|---------------|--------------|-------------|----------|--------------|--------------|-------------|----------|------------|
| 16:00 flat    | 1324         | +2.647      | 59.36    | \$1,478,384  | 325          | +2.481      | 54.76    | \$658,570  |
| Stop-loss hit | 2770         | -1.085      | 0.00     | -\$1,222,415 | 676          | -1.020      | 0.00     | -\$596,239 |

### Entry hour (New York)

| Entry hour  | Trades (all) | Avg R (all) | PF (all) | Net (all) | Trades 23-26 | Avg R 23-26 | PF 23-26 | Net 23-26 |
|-------------|--------------|-------------|----------|-----------|--------------|-------------|----------|-----------|
| 09:00-09:59 | 1297         | +0.325      | 1.43     | \$149,837 | 295          | +0.466      | 1.48     | \$73,785  |
| 10:00-10:59 | 857          | -0.021      | 1.06     | \$17,472  | 199          | -0.044      | 0.94     | -\$8,376  |
| 11:00-11:59 | 441          | +0.097      | 1.18     | \$28,136  | 107          | -0.011      | 1.09     | \$7,657   |
| 12:00-12:59 | 493          | +0.097      | 1.10     | \$18,768  | 126          | -0.027      | 0.94     | -\$5,254  |
| 13:00-13:59 | 414          | +0.015      | 1.33     | \$40,389  | 107          | -0.052      | 1.15     | \$9,162   |
| 14:00-14:59 | 401          | +0.033      | 1.06     | \$6,936   | 113          | +0.040      | 0.88     | -\$7,452  |
| 15:00-15:59 | 191          | -0.075      | 0.89     | -\$5,569  | 54           | -0.118      | 0.71     | -\$7,191  |

### Weekday

| Weekday | Trades (all) | Avg R (all) | PF (all) | Net (all) | Trades 23-26 | Avg R 23-26 | PF 23-26 | Net 23-26 |
|---------|--------------|-------------|----------|-----------|--------------|-------------|----------|-----------|
| Mon     | 818          | +0.103      | 1.21     | \$47,928  | 179          | -0.096      | 0.89     | -\$12,171 |
| Tue     | 853          | +0.276      | 1.37     | \$94,268  | 201          | +0.305      | 1.35     | \$40,051  |
| Wed     | 828          | -0.009      | 0.96     | -\$10,627 | 205          | +0.048      | 0.99     | -\$915    |
| Thu     | 855          | +0.122      | 1.25     | \$70,745  | 231          | +0.137      | 1.12     | \$18,211  |
| Fri     | 740          | +0.112      | 1.23     | \$53,655  | 185          | +0.167      | 1.16     | \$17,155  |

## H43 (#12, long + short): robustness

**Parameter neighbourhood.** Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

| Entry parameters              | Dir.  | Disc. n / PF | Conf. n / PF | Holdout n / PF | Holdout net | 2026 n / PF | 2026 net  |
|-------------------------------|-------|--------------|--------------|----------------|-------------|-------------|-----------|
| z=1.0, a=0.1                  | long  | 2876 / 1.02  | 1350 / 1.23  | 1093 / 1.07    | \$44,318    | 218 / 0.93  | -\$13,297 |
| z=1.0, a=0.1                  | short | 2883 / 1.03  | 1349 / 1.02  | 1030 / 1.03    | \$16,115    | 207 / 0.98  | -\$3,523  |
| z=1.0, a=0.1                  | both  | 4194 / 1.06  | 1943 / 1.18  | 1544 / 1.14    | \$116,639   | 301 / 1.06  | \$15,066  |
| z=1.5, a=0.1                  | long  | 2069 / 1.02  | 969 / 1.23   | 782 / 1.05     | \$23,347    | 168 / 0.97  | -\$4,402  |
| z=1.5, a=0.1                  | short | 2072 / 1.10  | 986 / 1.09   | 768 / 1.01     | \$5,053     | 153 / 0.95  | -\$7,752  |
| z=1.5, a=0.1                  | both  | 3250 / 1.08  | 1546 / 1.16  | 1239 / 1.11    | \$74,999    | 234 / 1.02  | \$3,759   |
| z=1.0, a=0.25                 | long  | 1349 / 1.03  | 621 / 1.27   | 531 / 1.22     | \$65,336    | 105 / 1.38  | \$29,960  |
| z=1.0, a=0.25                 | short | 1359 / 1.08  | 618 / 1.21   | 532 / 1.05     | \$14,312    | 112 / 0.84  | -\$16,268 |
| <b>z=1.0, a=0.25 ◀ frozen</b> | both  | 2123 / 1.16  | 970 / 1.39   | 836 / 1.14     | \$64,961    | 165 / 0.98  | -\$2,630  |
| z=1.5, a=0.25                 | long  | 956 / 0.95   | 436 / 1.25   | 374 / 1.14     | \$31,409    | 72 / 1.00   | \$77      |
| z=1.5, a=0.25                 | short | 984 / 1.08   | 472 / 1.24   | 378 / 1.18     | \$40,008    | 82 / 0.57   | -\$36,658 |
| z=1.5, a=0.25                 | both  | 1639 / 1.09  | 788 / 1.19   | 631 / 1.11     | \$41,226    | 124 / 0.63  | -\$42,781 |
| z=1.0, a=None                 | long  | 5638 / 0.94  | 2626 / 1.04  | 1975 / 1.08    | \$94,195    | 397 / 1.03  | \$12,812  |
| z=1.0, a=None                 | short | 5360 / 0.94  | 2450 / 1.04  | 1896 / 0.97    | -\$30,959   | 363 / 1.06  | \$17,918  |
| z=1.0, a=None                 | both  | 7490 / 0.93  | 3407 / 1.05  | 2567 / 1.11    | \$151,907   | 511 / 1.06  | \$27,881  |
| z=1.5, a=None                 | long  | 4148 / 0.94  | 1974 / 1.03  | 1486 / 1.01    | \$8,696     | 323 / 0.95  | -\$16,422 |
| z=1.5, a=None                 | short | 3840 / 1.00  | 1778 / 1.13  | 1361 / 1.05    | \$42,141    | 285 / 0.99  | -\$2,585  |
| z=1.5, a=None                 | both  | 5842 / 0.99  | 2698 / 1.09  | 2033 / 1.10    | \$115,318   | 421 / 0.93  | -\$27,374 |

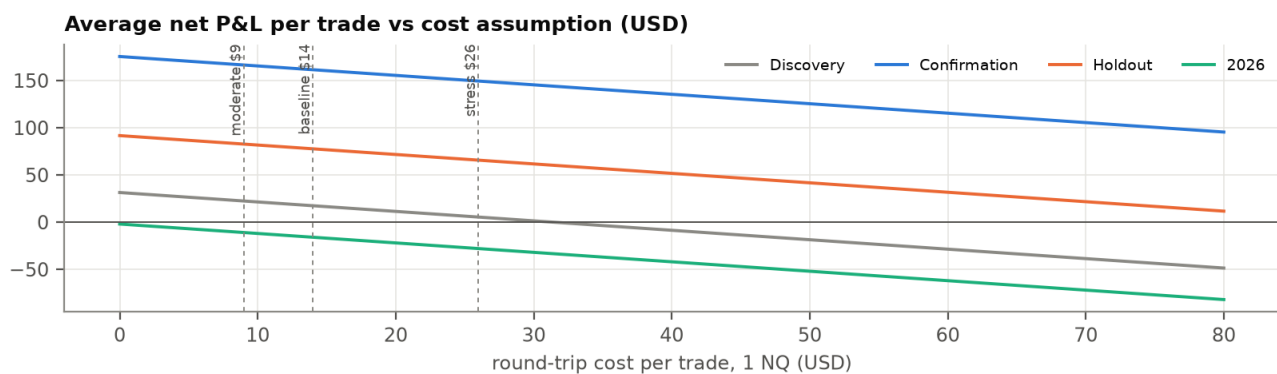
Neighbours with  $\geq 20$  trades and positive net: Discovery 11/18; Confirmation 18/18; Holdout 17/18; 2026 7/18

**Stop / exit management.** The frozen entry under every management variant from the campaign library (discovery stages A and B).

| Stop / exit management                               | Disc. PF | Conf. PF | Holdout n / PF | Holdout net | 2026 PF | 2026 net  |
|------------------------------------------------------|----------|----------|----------------|-------------|---------|-----------|
| 1×ATR stop; single target at +1R                     | 0.87     | 1.15     | 1309 / 1.09    | \$46,204    | 0.95    | -\$8,451  |
| 1×ATR stop; single target at +1.5R                   | 0.98     | 1.22     | 1189 / 1.11    | \$63,254    | 1.04    | \$7,075   |
| 1×ATR stop; single target at +2R                     | 1.03     | 1.16     | 1086 / 1.08    | \$42,141    | 0.88    | -\$20,603 |
| 1×ATR stop; single target at +3R                     | 1.07     | 1.27     | 982 / 1.09     | \$49,382    | 1.06    | \$9,948   |
| 1×ATR stop; single target at +4R                     | 1.13     | 1.25     | 931 / 1.11     | \$57,621    | 1.13    | \$19,144  |
| <b>1×ATR stop; no target, flat 16:00 ◀ frozen</b>    | 1.16     | 1.39     | 836 / 1.14     | \$64,961    | 0.98    | -\$2,630  |
| 1×ATR stop; no target, exit after 60 min             | 0.94     | 1.28     | 1055 / 1.14    | \$64,905    | 1.03    | \$4,316   |
| 1.5×ATR stop; single target at +1R                   | 1.03     | 1.25     | 1067 / 1.08    | \$45,607    | 1.05    | \$8,898   |
| 1.5×ATR stop; single target at +2R                   | 1.11     | 1.29     | 877 / 1.11     | \$62,112    | 1.01    | \$1,189   |
| 1.5×ATR stop; no target, flat 16:00                  | 1.17     | 1.42     | 737 / 1.14     | \$68,157    | 0.93    | -\$10,985 |
| 5-min swing stop; single target at +1R               | 1.01     | 1.27     | 879 / 0.94     | -\$36,611   | 1.06    | \$9,936   |
| 5-min swing stop; single target at +2R               | 1.07     | 1.27     | 748 / 0.97     | -\$20,157   | 0.99    | -\$1,419  |
| 5-min swing stop; no target, flat 16:00              | 1.07     | 1.39     | 675 / 1.02     | \$11,955    | 1.08    | \$12,133  |
| 1×ATR stop; stop to entry at +0.5R, target +2R       | 0.62     | 1.06     | 1317 / 0.79    | -\$82,458   | 0.62    | -\$53,613 |
| 1×ATR stop; stop to entry at +0.5R, no target        | 0.71     | 1.25     | 1173 / 0.88    | -\$43,682   | 0.42    | -\$77,409 |
| 1×ATR stop; stop to entry at +1R, target +2R         | 0.91     | 1.17     | 1146 / 1.02    | \$7,616     | 0.84    | -\$24,410 |
| 1×ATR stop; stop to entry at +1R, no target          | 0.97     | 1.38     | 962 / 1.07     | \$28,167    | 0.64    | -\$44,463 |
| 1×ATR stop; stop to entry at +1.5R, target +2R       | 1.01     | 1.24     | 1111 / 1.10    | \$49,906    | 0.89    | -\$19,474 |
| 1×ATR stop; stop to entry at +1.5R, no target        | 1.08     | 1.54     | 910 / 1.12     | \$50,645    | 0.70    | -\$40,776 |
| 1×ATR stop; no target, 1.5×ATR trailing stop         | 0.98     | 1.27     | 1042 / 1.17    | \$74,262    | 0.86    | -\$19,554 |
| 1×ATR stop; no target, 2×ATR trailing stop           | 1.00     | 1.36     | 952 / 1.09     | \$39,832    | 0.79    | -\$31,058 |
| 1×ATR stop; no target, 3×ATR trailing stop           | 1.16     | 1.40     | 876 / 1.14     | \$66,421    | 0.97    | -\$4,506  |
| 1×ATR stop; no target, trail beyond last 5-min swing | 1.03     | 1.28     | 967 / 1.09     | \$40,292    | 0.99    | -\$1,454  |
| 1×ATR stop; 50% at +1R, 50% at +2R                   | 0.97     | 1.19     | 1086 / 1.06    | \$27,904    | 0.90    | -\$13,236 |
| 1×ATR stop; 50% at +1R, 50% at +3R                   | 0.98     | 1.25     | 982 / 1.07     | \$29,614    | 1.08    | \$9,246   |
| 1×ATR stop; 50% at +1R, rest trails 2×ATR            | 0.94     | 1.30     | 951 / 1.05     | \$18,891    | 0.88    | -\$14,906 |
| 1×ATR stop; 25% at +1R, 75% runs to 16:00            | 1.11     | 1.38     | 836 / 1.11     | \$45,018    | 0.98    | -\$1,984  |

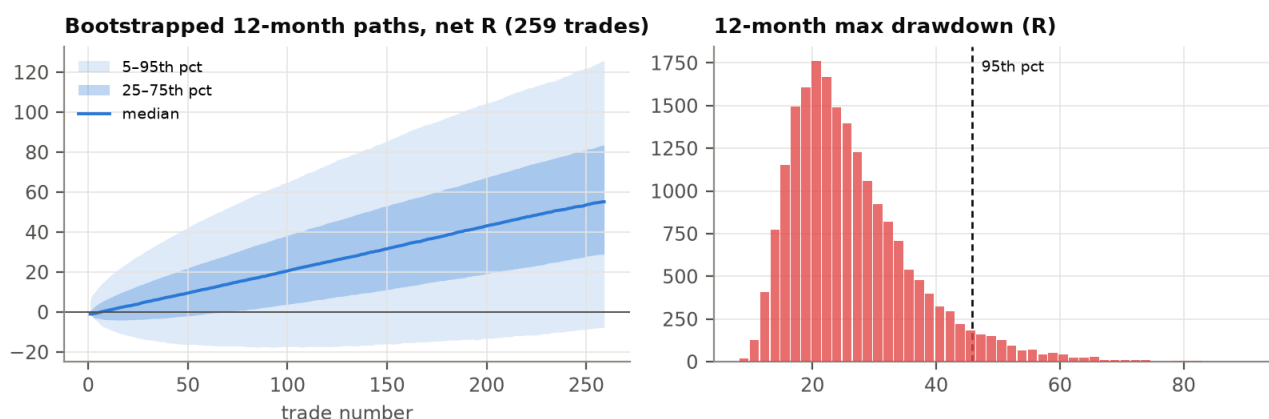
| Stop / exit management                              | Disc. PF | Conf. PF | Holdout n / PF | Holdout net | 2026 PF | 2026 net  |
|-----------------------------------------------------|----------|----------|----------------|-------------|---------|-----------|
| 1×ATR stop; 50% at +1R then stop to entry, rest +3R | 0.89     | 1.17     | 1077 / 1.05    | \$23,024    | 0.95    | −\$7,462  |
| 1×ATR stop; stop to entry at +1R, 2×ATR trail       | 0.90     | 1.29     | 1010 / 1.08    | \$31,820    | 0.77    | −\$29,079 |

**Cost sensitivity.** The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$31; Confirmation \$175; Holdout \$92; 2026 n/a (gross loss). The baseline assumption is \$14.

## H43 (#12, long + short): risk, sizing and forward expectations



Bootstrap of 1971 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 259 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown. It assumes the 2019–26 edge continues unchanged; the section 'Will it perform like this in the future?' repeats it with smaller edges.

| Forward 12-month statistic (bootstrap) | In R   | 1 NQ at 2026 risk (1R ≈ \$1,293) | 1 MNQ (1R ≈ \$129) |
|----------------------------------------|--------|----------------------------------|--------------------|
| Median 12-month result                 | +55.2  | \$71,305                         | \$7,130            |
| 5th percentile (bad year)              | −7.8   | −\$10,032                        | −\$1,003           |
| 95th percentile (good year)            | +125.8 | \$162,563                        | \$16,256           |
| Probability of a losing year           | 7%     |                                  |                    |
| Median max drawdown                    | 24.2   | \$31,293                         | \$3,129            |
| 95th pct max drawdown                  | 45.8   | \$59,149                         | \$5,915            |
| 99th pct max drawdown                  | 59.2   | \$76,581                         | \$7,658            |
| Historical max drawdown 2010–26        | 123.3  |                                  |                    |

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

### Portfolio context (25-strategy frozen cohort, 1 contract each)

| Period  | Corr. with rest of cohort (daily) | Median pairwise corr. | Most correlated members                                                            | Share of cohort P&L |
|---------|-----------------------------------|-----------------------|------------------------------------------------------------------------------------|---------------------|
| holdout | 0.47                              | 0.12                  | #1 H20 both (0.50), #19 H41 both (0.40), #7 H07 both (0.21), #16 H50 short (0.21)  | 10%                 |
| 2026    | 0.11                              | 0.04                  | #22 H47 long (0.22), #4 H47 long (0.21), #25 H07 short (0.20), #10 H08 long (0.20) | -3%                 |

# Strengths and weaknesses

## Strengths

- **Best search-corrected evidence among the four.** Confirmation  $t$  3.27 (3.79 in R) and family-wise  $p$  0.020. The result held up even after accounting for everything searched.
- **Positive in R every year since 2018.** That is nine calendar years in a row, including all four post-freeze years (+37R, +20R, +34R, +26R). For a fixed-risk trader this is the relevant record.
- **Fixed risk improves it.** 2019–26 PF is about 1.28 at fixed risk against 1.22 on fixed contracts, because its weak trades are the big-stop, high-volatility ones.
- **Robust to exits and parameters before 2026.** All 12 filtered grid neighbours were profitable in the holdout, and 25 of 29 stop/exit schemes were too.
- **Large sample and fast feedback.** About 277 trades a year. 150 forward trades take about 7 months.
- **Costs are not the issue.** The stress-cost PF after the freeze is 1.08, only slightly below the baseline 1.10.

## Weaknesses and risks

- **Marginal holdout, fading edge.** Holdout  $t$  1.15 and null  $p$  0.079. The average trade fell from +0.32R in confirmation to +0.11R in the holdout. The post-freeze  $t$  is 1.55 in R.
- **The filter's advantage disappeared after the freeze.** Unfiltered 15-minute momentum did about as well in 2023–26. If the filter no longer matters, H43 is largely generic momentum, and that is already covered by H41.
- **Overlap with H41 and H20.** Holdout daily correlation is 0.40 with H41, 0.50 with H20, and 0.47 with the rest of the cohort. It adds momentum risk rather than a new source of return.
- **The edge was absent for years.** 2010–17 made –57R in total, and the discovery drawdown lasted 2,295 days. Like H41, the effect appeared from 2018.
- **Concentrated edge.** Longs made +0.20R per trade, shorts only +0.04R. The first hour carries most of the profit. Calm days made +0.28R, mid- and high-volatility days about zero.
- **2026 dollar loss and wide stops.** On 1 NQ, 2026 was –\$2.6k with a \$32.5k drawdown, and the stop is now about 60 points, so a strict \$150 MNQ budget skips about a quarter of signals.
- **Busy days.** Up to 7–9 trades in one session, and long losing runs of 19 trades in every period before 2026.

## Will it perform like this in the future?

**Short answer: probably weakly positive in R, with low confidence.** For: strong search-corrected confirmation, nine straight positive years in R, and a sensible mechanism. Against: the holdout edge was a third of confirmation's, the filter stopped mattering, and it was picked fourth of 25 after the holdout. The table resamples its own trades at five assumed average trades.

| Scenario for the next 12 months        | Avg R / trade | PF (R) | Median 12-mo | Bad year (5th pct) | Good year (95th) | P(losing year) | 95th pct max DD |
|----------------------------------------|---------------|--------|--------------|--------------------|------------------|----------------|-----------------|
| As in the post-freeze record (2023–26) | +0.117        | 1.15   | +29R         | –34R               | +99R             | 23%            | 61R             |
| As in confirmation (2019–22)           | +0.324        | 1.52   | +83R         | +20R               | +153R            | 1%             | 35R             |
| Planning case: edge roughly halved     | +0.060        | 1.07   | +14R         | –49R               | +85R             | 36%            | 71R             |
| Weak edge                              | +0.030        | 1.04   | +7R          | –57R               | +77R             | 43%            | 76R             |
| Edge gone                              | +0.000        | 1.00   | –1R          | –64R               | +69R             | 51%            | 82R             |

*Each row resamples the 2019–2026 trades (259 trades per simulated year) with the average trade shifted to the scenario value. In dollars, multiply R by about \$1,293 for 1 NQ or \$129 for 1 MNQ at 2026 volatility. The highlighted row is the planning case this report recommends.*

**Our planning case is the highlighted row: about +0.06R per trade**, half the post-freeze average. With about 260 trades a year that gives a median year of about +14R, but roughly a 36% chance of a losing year and 12-month drawdowns of 70R at the 95th percentile. The edge per trade is small next to the noise.

**What would make us more confident:** 150+ forward trades at +0.08R or better, and the filtered version beating unfiltered momentum again. **What would make us drop it:** 300 forward trades averaging below 0R, or a drawdown beyond 70R.

# Fixed risk on MNQ instead of fixed contracts

**Bottom line.** Fixed risk **helps** H43, as it did H41. Over 2019–26 the profit factor rises from 1.22 on 1 NQ to about 1.28. Its edge sits in the calmer days, and fixed contracts put most of their dollars on the volatile ones. The catch, again like H41: in 2026 the median stop was 60 points = \$120 per MNQ, so a strict \$150 rule skips 27% of 2026 signals. Use the same '1 MNQ up to \$250' rule as H41. Because it is the same kind of trade as H41 (intraday momentum), it adds much less to an H41 account than H47 or H25 did.

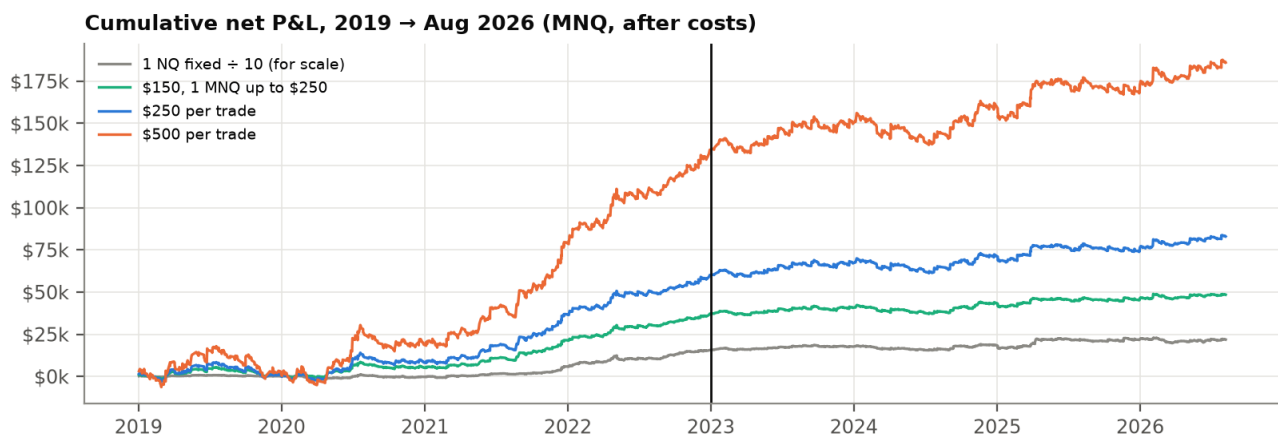
## What changes

- **Sizing rule.** Contracts = risk budget ÷ (stop distance in points × \$2), rounded down. With the cap rule, take 1 MNQ when that risks no more than \$250, otherwise skip. MNQ costs are assumed at \$2.50 per round trip (\$4.25 stress).
- **Why the profit factor goes up.** Over 2019–26, trades on the calmest third of days (stop small relative to price) averaged +0.50R. The middle and most volatile thirds averaged +0.12R and +0.03R. Fixed risk stops the weak, high-volatility trades from dominating the dollar result. That is also why 2026 was −\$2.6k on 1 NQ but +26R per unit of risk.
- **Frequent trades, clustered losses.** It trades about 5 times a week and up to 7–9 times in one day. With the cap rule, the worst post-freeze day was −\$1,122 and the worst drawdown \$5,151. At \$250 per trade the worst day was −\$1,396. Check this against your daily loss limit.

## Post-freeze 2023 → Aug 2026 (data never used to select the strategy)

| Metric                       | 1 NQ fixed      | 1 MNQ fixed  | \$100 risk   | \$150 skip if 0 | \$150, 1 MNQ up to \$250 | \$250 risk    | \$500 risk    |
|------------------------------|-----------------|--------------|--------------|-----------------|--------------------------|---------------|---------------|
| Trades taken (skipped)       | 1001 (0)        | 1001 (0)     | 687 (314)    | 900 (101)       | 990 (11)                 | 990 (11)      | 1001 (0)      |
| Profit factor                | 1.10            | 1.08         | 1.20         | 1.17            | 1.13                     | 1.16          | 1.16          |
| Profit factor, stress costs  | 1.08            | 1.05         | 1.16         | 1.13            | 1.09                     | 1.12          | 1.12          |
| Win rate                     | 29%             | 28%          | 27%          | 27%             | 28%                      | 28%           | 28%           |
| Net P&L                      | \$62,331        | \$5,132      | \$7,860      | \$12,636        | \$11,288                 | \$22,917      | \$51,787      |
| Net P&L per year             | \$17,266        | \$1,422      | \$2,177      | \$3,500         | \$3,127                  | \$6,348       | \$14,345      |
| Average trade                | \$62            | \$5          | \$11         | \$14            | \$11                     | \$23          | \$52          |
| Average \$ risk per trade    | \$905           | \$91         | \$77         | \$116           | \$122                    | \$206         | \$457         |
| Risk range (min - max)       | \$190 - \$3,540 | \$19 - \$354 | \$50 - \$100 | \$76 - \$150    | \$76 - \$242             | \$126 - \$250 | \$270 - \$500 |
| Contracts (avg / max)        | 1.0 / 1         | 1.0 / 1      | 1.3 / 5      | 1.8 / 7         | 1.7 / 7                  | 3.0 / 13      | 6.5 / 26      |
| Max drawdown (closed trades) | \$32,482        | \$3,424      | \$2,700      | \$5,151         | \$5,151                  | \$8,628       | \$18,794      |
| Worst losing run             | −\$20,986       | −\$2,120     | −\$1,374     | −\$2,122        | −\$2,538                 | −\$3,876      | −\$8,638      |
| Worst day                    | −\$11,148       | −\$1,122     | −\$467       | −\$693          | −\$1,122                 | −\$1,396      | −\$3,118      |

1 NQ = the frozen strategy exactly. MNQ columns use \$2/point and \$2.50 per round trip per contract. Drawdowns use closed-trade equity.



## Full out-of-sample record 2019 → Aug 2026

| Metric                       | 1 NQ fixed     | 1 MNQ fixed  | \$100 risk   | \$150 skip if 0 | \$150, 1 MNQ up to \$250 | \$250 risk    | \$500 risk    |
|------------------------------|----------------|--------------|--------------|-----------------|--------------------------|---------------|---------------|
| Trades taken (skipped)       | 1971 (0)       | 1971 (0)     | 1490 (481)   | 1831 (140)      | 1957 (14)                | 1957 (14)     | 1971 (0)      |
| Profit factor                | 1.22           | 1.19         | 1.34         | 1.32            | 1.28                     | 1.28          | 1.28          |
| Profit factor, stress costs  | 1.19           | 1.15         | 1.27         | 1.26            | 1.22                     | 1.22          | 1.23          |
| Win rate                     | 30%            | 29%          | 28%          | 29%             | 29%                      | 29%           | 29%           |
| Net P&L                      | \$218,956      | \$19,728     | \$29,568     | \$50,832        | \$48,477                 | \$83,010      | \$186,057     |
| Net P&L per year             | \$28,772       | \$2,592      | \$3,885      | \$6,680         | \$6,370                  | \$10,908      | \$24,449      |
| Average trade                | \$111          | \$10         | \$20         | \$28            | \$25                     | \$42          | \$94          |
| Average \$ risk per trade    | \$772          | \$77         | \$79         | \$120           | \$124                    | \$213         | \$463         |
| Risk range (min - max)       | \$95 - \$3,540 | \$10 - \$354 | \$50 - \$100 | \$76 - \$150    | \$76 - \$248             | \$126 - \$250 | \$254 - \$500 |
| Contracts (avg / max)        | 1.0 / 1        | 1.0 / 1      | 1.8 / 10     | 2.5 / 15        | 2.4 / 15                 | 4.3 / 26      | 9.0 / 52      |
| Max drawdown (closed trades) | \$32,482       | \$3,424      | \$3,948      | \$6,258         | \$6,258                  | \$10,853      | \$23,038      |
| Worst losing run             | -\$24,110      | -\$2,428     | -\$1,374     | -\$2,122        | -\$2,538                 | -\$4,027      | -\$8,892      |
| Worst day                    | -\$16,916      | -\$1,702     | -\$476       | -\$778          | -\$1,443                 | -\$1,635      | -\$3,720      |

## Stop size and contracts by year

| Year | Trades | Median stop (pts) | 1 MNQ risk | \$100: MNQ (skip) | \$150: MNQ (skip) | \$200: MNQ (skip) | \$300: MNQ (skip) | \$500: MNQ (skip) |
|------|--------|-------------------|------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 2019 | 234    | 12.0              | \$24       | 4 (0%)            | 6 (0%)            | 8 (0%)            | 12 (0%)           | 20 (0%)           |
| 2020 | 248    | 28.8              | \$58       | 1 (16%)           | 2 (5%)            | 3 (2%)            | 5 (0%)            | 8 (0%)            |
| 2021 | 221    | 25.5              | \$51       | 2 (9%)            | 2 (0%)            | 3 (0%)            | 5 (0%)            | 9 (0%)            |
| 2022 | 267    | 45.5              | \$91       | 1 (40%)           | 1 (10%)           | 2 (3%)            | 3 (0%)            | 5 (0%)            |
| 2023 | 264    | 30.8              | \$62       | 1 (8%)            | 2 (1%)            | 3 (0%)            | 4 (0%)            | 8 (0%)            |
| 2024 | 283    | 34.8              | \$70       | 1 (18%)           | 2 (2%)            | 2 (0%)            | 4 (0%)            | 7 (0%)            |
| 2025 | 289    | 47.2              | \$94       | 1 (45%)           | 1 (16%)           | 2 (4%)            | 3 (1%)            | 5 (0%)            |
| 2026 | 165    | 59.8              | \$120      | 1 (68%)           | 1 (27%)           | 1 (10%)           | 2 (1%)            | 4 (0%)            |

MNQ = median contracts per trade at that budget; skip = share of signals where 1 MNQ risks more than the budget.

## Prop-firm evaluation

Each simulated evaluation draws random whole trading days from 2019-Aug 2026 (every trade of that day, in time order) for up to 250 trading days. It passes at the target and fails at the trailing drawdown (closed trades) or the daily loss limit. 5,000 runs per row. The rules are generic examples. The reference account runs H41 (\$150, 1 MNQ up to \$250), H47 (\$150) and H25 short (\$100).

| Account set-up                                                              | Pass | Fail | Not done in 250 days | Median days to pass |
|-----------------------------------------------------------------------------|------|------|----------------------|---------------------|
| <b>Example A: target \$3,000 · trailing DD \$2,000 · daily loss \$1,000</b> |      |      |                      |                     |
| H43 alone, \$150 (1 MNQ up to \$250)                                        | 50%  | 50%  | 0%                   | 46                  |
| H43 alone, \$250 per trade                                                  | 40%  | 60%  | 0%                   | 18                  |
| H43 alone, \$500 per trade                                                  | 31%  | 69%  | 0%                   | 5                   |
| H41 + H47 + H25 (reference, no H43)                                         | 60%  | 40%  | 0%                   | 31                  |
| H41 + H47 + H25 + H43 at \$100                                              | 56%  | 44%  | 0%                   | 21                  |
| H41 + H47 + H25 + H43 at \$150 (1 MNQ up to \$250)                          | 49%  | 51%  | 0%                   | 15                  |
| <b>Example B: target \$6,000 · trailing DD \$3,000 · daily loss \$2,000</b> |      |      |                      |                     |
| H43 alone, \$150 (1 MNQ up to \$250)                                        | 49%  | 44%  | 7%                   | 118                 |
| H43 alone, \$250 per trade                                                  | 40%  | 60%  | 0%                   | 48                  |
| H43 alone, \$500 per trade                                                  | 31%  | 69%  | 0%                   | 13                  |
| H41 + H47 + H25 (reference, no H43)                                         | 67%  | 32%  | 1%                   | 80                  |
| H41 + H47 + H25 + H43 at \$100                                              | 60%  | 40%  | 0%                   | 55                  |
| H41 + H47 + H25 + H43 at \$150 (1 MNQ up to \$250)                          | 52%  | 48%  | 0%                   | 40                  |

- **Alone it behaves like a weaker H41.** At \$150 (1 MNQ up to \$250) it passes 50% of Example A runs, with a median of 46 days.
- **As a fourth sleeve it lowers the pass rate.** The three-strategy account passes 60% (Example A, median 31 days) and 67% (Example B, 80 days). Adding H43 at just \$100 gives 56% and 60%. It is faster (21 and 55 days), but mainly because the account is now taking more momentum risk. It overlaps H41 (holdout daily correlation 0.40), and its trades often lose on the same days, so it uses up the drawdown cushion.

- **Suggested use:** do not add H43 to the evaluation account. Keep it on paper (the forward record). If you trade it later, fold it into H41's momentum budget instead of adding risk, for example by splitting one \$150-\$200 momentum budget between the two.
- As before, many firms trail on unrealised equity (H43 holds runners to 15:59), and the day-resampling does not capture long bad regimes such as 2010-17, when this edge was absent.

## Discussion

**Strength of evidence.** Over 2010–2026: +500R on 4,094 trades (+0.12R per trade,  $t$  3.11 in R, PF 1.21 in dollars). Almost all of it came after 2017. The post-freeze part is 1,001 trades at +0.12R ( $t$  1.55). It is positive, but the untouched evidence is thin relative to the number of trades.

**Reading 2026.** –\$2.6k on 1 NQ but +26R. The losses came on the largest-stop, most volatile days, where one NQ risked two to three times the average. Fixed-risk sizing, which you use, would have been positive.

**Where the P&L comes from.** All of it comes from the trades held to the close. Across all years, the long side made four times the short side per trade. First-hour entries and calm-day entries carry the edge. None of this should change the rules. It does show that in practice H43 is mostly 'morning momentum, long-leaning, in calm trending markets'.

**Management.** Before 2026 nearly every stop/exit scheme was profitable (25 of 29 in the holdout). In 2026 most schemes lost in dollars, which matches the dollar-vs-R pattern above rather than a change in the entry signal.

**Portfolio fit.** The three recommended strategies cover different ground: all-day momentum (H41), a 09:45 gap-and-go long (H47) and Nasdaq-specific weakness short (H25). H43 is a second momentum strategy on top of H41. Its best use is as a replacement or split of H41's momentum budget, not an addition.

## Recommendation

**Recommendation: FOURTH CANDIDATE: FORWARD-PAPER ONLY FOR NOW. Do not add it to the evaluation account.** It has the strongest search-corrected confirmation result of the four candidates ( $t$  3.27, family-wise  $p$  0.020), and it has been positive in R every year since 2018, including every year after the freeze. But its untouched holdout was only marginal (PF 1.14,  $t$  1.15, null  $p$  0.079), and 2026 lost money on fixed contracts. It is the same kind of trade as H41 (holdout daily correlation 0.40). In the simulations, adding it to an H41 + H47 + H25 account *lowered* the pass rate.

**Decision requested:** keep H43 in the true-forward **paper** record. Do not trade it in the evaluation account.

**If you want to trade it later:** MNQ at \$100–\$150 fixed risk with the '1 MNQ up to \$250' rule, and only by splitting H41's momentum budget (e.g. \$100 each instead of \$150–\$200 on H41 alone), never on top of it.

**Re-evaluate after about 7 months** (about 150 forward trades), together with H41's first gate.

### Conditions, limits and kill criteria

- Promotion gate: at least 150 true-forward trades averaging +0.08R or better, net positive under stress costs, and the filtered signal outperforming the unfiltered momentum trigger over the same period.
- Drop rule: 300 forward trades averaging below 0R, or a drawdown beyond 70R (the planning-case 95th percentile).
- Budget: never size H43 and H41 independently. Their combined momentum risk should not exceed what H41 alone would carry.
- No parameter, filter or management changes. Every observation in this report is descriptive.

### Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.
- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null  $p$  0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the strategy. The preregistered process says the only valid new evidence now comes from the true-forward record starting 24 Sep 2026.
- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

# Appendix A: frozen specification (verbatim)

## H43 (#12, long + short)

```
{
  "costs": {
    "baseline_round_trip_usd": 14.0,
    "cost_version": "costs_v2_approved_2026-09-21#a27ed5df0c04"
  },
  "direction": "both",
  "entry_conditions": {
    "down_event": "the 15-minute move first falls below -1.0 sigma units with the 5-minute return autocorrelation (last 24) >= 0.25",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=both",
    "up_event": "the 15-minute move first exceeds +1.0 sigma units with the 5-minute return autocorrelation (last 24) >= 0.25"
  },
  "entry_id": "E4de4e4fa7bbaa856",
  "family": "v3_h43_pos_autocorr_momentum",
  "instrument": "NQ",
  "management": {
    "abs_target": null,
    "be_R": null,
    "be_after_leg": null,
    "exit_opp": false,
    "hold": "eod",
    "id": "STOP_TIME_eod",
    "kind": "stop_time",
    "legs": [
      [
        1.0,
        null
      ]
    ],
    "needs_stop": true,
    "params": {
      "hold": "eod"
    },
    "trail": null
  },
  "parameters": {
    "a": 0.25,
    "mode": "continuation",
    "z": 1.0
  },
  "stop": {
    "method": "atr",
    "mult": 1.0
  },
  "strategy_id": "C16cd8e3c605af462",
  "trading_window": {
    "entry_end": "15:30",
    "entry_start": "09:30",
    "flat_at": "16:00",
    "name": "rth"
  }
}
```

## Appendix B: provenance

| Item                          | Value                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final cohort freeze           | freezes/FINAL_COHORT_FREEZE.json · sha256<br>f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03                                                                                |
| Frozen (UTC)                  | 2026-09-22T23:21:51.429326+00:00                                                                                                                                                             |
| Discovery freeze              | sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9                                                                                                                      |
| Master record                 | results/V3_MASTER_RESEARCH_RECORD.md                                                                                                                                                         |
| Sealed holdout / audit trades | results/holdout/trades.parquet · results/audit2026/trades.parquet                                                                                                                            |
| This report                   | reports/H43_Autocorrelation_Momentum_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence |
| Known erratum                 | results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)                                                                                                      |